BOARDGAME ARE MADE OF RULES

Riccardo Conforto Galli

INTRODUCTION

The aim of this project is to evaluate how well different large language models process analyze and extract information from board-game rule texts.

DATA

- 5 board games in pdf format
- from BGG Hall of Fame
- complexity from light to medium

GAME	BGG-id
7 Wonders	316377
Catan	13
Dominion	209418
Power Grid	2651
Ticket to Ride	9209

RESEARCH QUESTION

1. Ask the model to explain the game to different kind of audiences

2. Feed the model a version of a game with missing or flawed rules

3. Give the model a rule book and ask it to classify the mechanics and estimate the parameters

MODELS

run locally

temperature set to the recommended value if present, 0.7 otherwise

1. LLaMA 3.1 8b
2. Qwen 2.5 7b
3. Gemma 3n 4b

EXPLANATION

prompted to generate an explanation in conversation terms for subjects of age 7, 11 and 16 years.

Evaluated on:

1. Readability
2. Completeness
3. Coinciseness

1. READABILITY

Three readability metric to evaluated generated outputs

1. Flesch-Kincaid
2. SMOG
3. Dale-Chall

Model	Age	Expected	Flesch–Kincaid	SMOG	Dale-Chall grade	Dale–Chall
LLaMA	7	2	4.68	7.72	10-12	8.25
	11	6	5.76	8.47	10-12	8.46
	16	11	6.31	9.12	12	8.99
Qwen	7	2	4.41	7.09	10-12	8.13
	11	6	4.70	7.58	10-12	8.56
	16	11	5.35	8.25	>12	9.31
Gemma	7	2	3.68	6.97	9-10	7.70
	11	6	4.41	7.81	10	8.01
	16	11	4.81	8.08	10-12	8.36

2. COMPLETENESS

1. Extraction of rule hierarchy
2. hierarchy restructuration
3. detect rule presence

Completeness score

$$\frac{\sum_{r \in R} \mathbb{I}_E(r) \cdot v(r)}{\sum_{r \in R} v(r)}$$

$$v(r) = 4 - h(r)$$

detect rule presence

Sentence embedding applied to sliding windows.
Detected if cosine similarity > 0.4

Hey there, young adventurer! Let's talk about the amazing game of Catan! **What's the goal?** The goal of the game is to collect 10 "victory points" (VPs). You earn VPs by building roads, settlements, and cities on the island of Catan. **How do you win?** You win the game when you have 10 or more VPs at the end of your turn. You can earn VPs by building settlements, cities, and roads, or by collecting special cards called "development cards". **What happens in a turn?** In a turn, you do two things: 1. **Production phase**: You roll two dice to see which hexes on the island will produce resources (like brick, wood, or wheat). If you have a settlement or city on a hex that produces resources, you get to collect those resources! 2. **Action phase**: You can use your resources to build roads, settlements, cities, or trade with other players. You can also play development cards to get special powers. **Some cool things to remember:** * When you roll a 7, you have to discard some of your resources and move the "robber" to a new hex. The robber will steal resources from another player who has a building on that hex! * You can trade resources with other players, but you can't give away cards you don't have. * When you build a road, settlement, or city, you have to follow some rules, like not building too close to other players' buildings. That's it! Catan is a super fun game that's all about building and trading. I hope you're excited to start playing!

sliding window representation for: "Discard Resources on a Roll of 7"

3. COINCISENESS

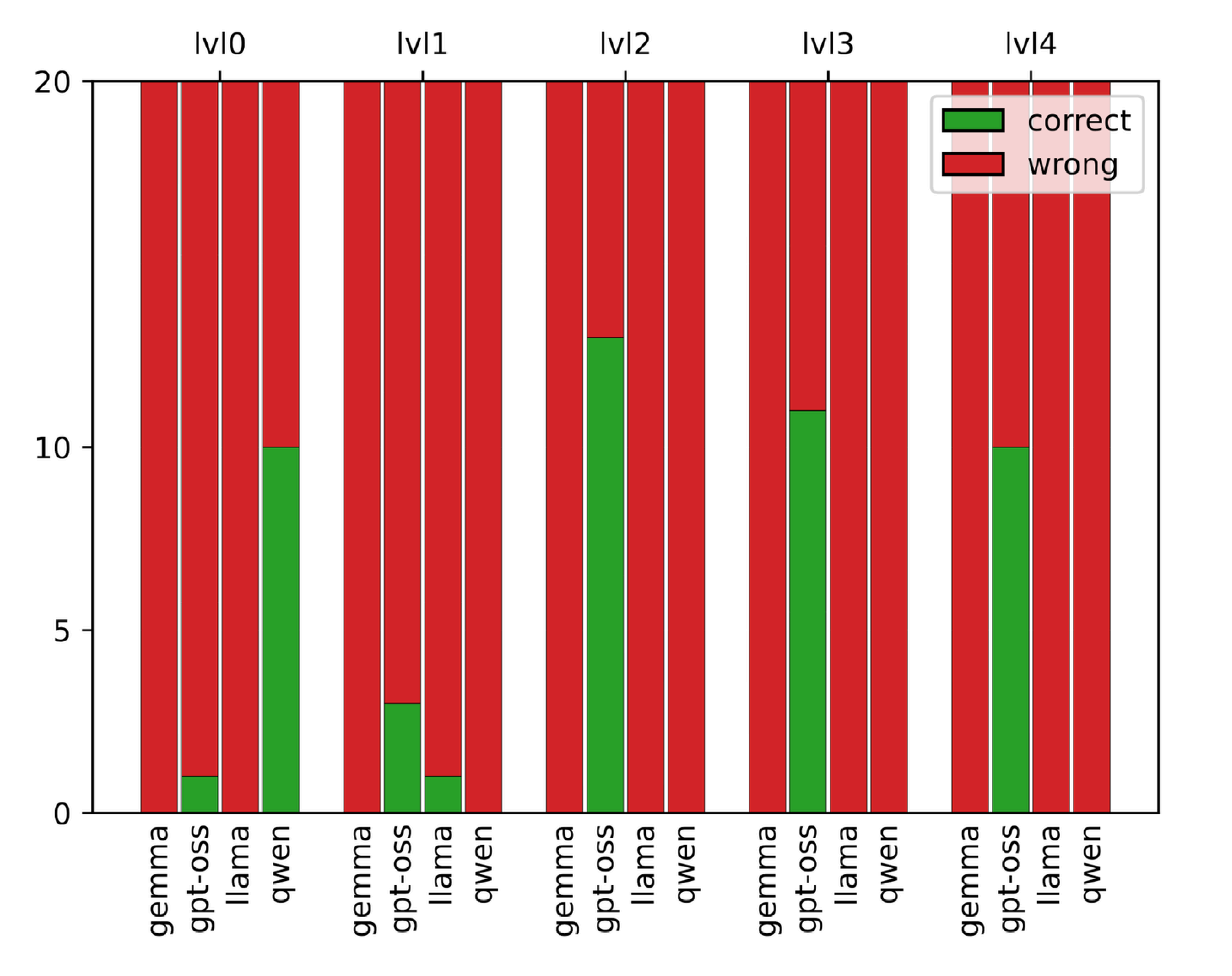
$$\frac{len(explanation)}{len(original)}$$

Game	**Model**	**Rule coverage ↑**	**Relative Size ↓**
7 Wonders	gemma	0.09	10.13
	llama	0.10	9.90
	qwen	0.07	10.68
Catan	gemma	0.17	11.86
	llama	0.18	12.28
	qwen	0.20	14.55
Dominion	gemma	0.39	6.61
	llama	0.36	6.75
	qwen	0.40	9.91
Power Grid	gemma	0.11	5.68
	llama	0.11	6.22
	qwen	0.10	5.50
Ticket to Ride	gemma	0.76	16.38
	llama	0.91	19.59
	qwen	0.90	23.32

ERROR DETECTION

five levels of edits:

0. Original
1. Missing
2. unsolvable
3. hard lock
4. unbalanced



lvl0
lvl1
lvl2
lvl3
lvl4
20
10
5
0
correct
wrong
gemma
gpt-oss
llama
qwen

PARAMETER ESTIMATION

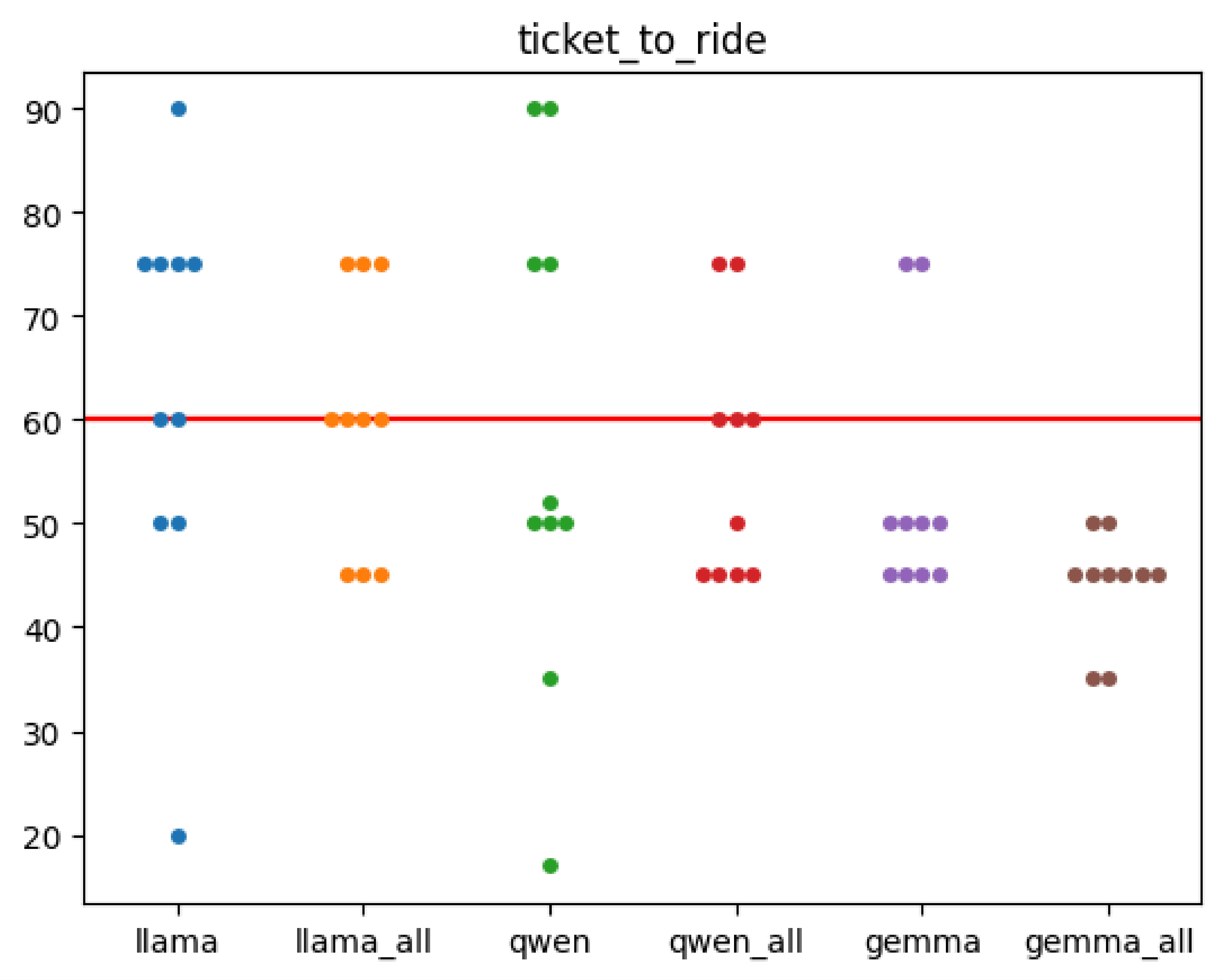
two different prompts:

- specific to each parameter
- all parameters at once

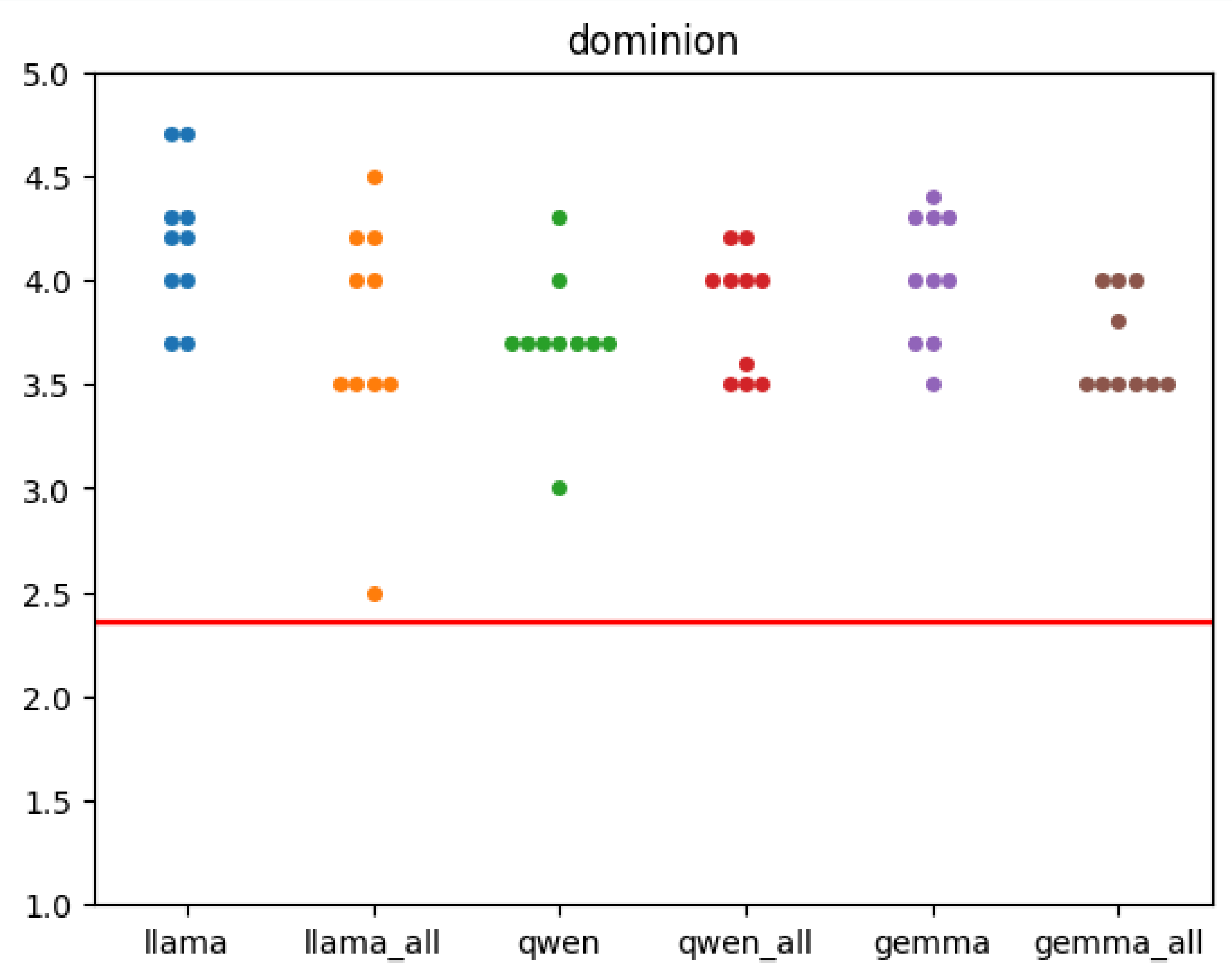
	Exact match		Loose match	
	mean precision	**mean recall**	**mean precision**	**mean recall**
gemma	0.144770	0.276071	0.256993	0.528810
llama	0.105373	0.119722	0.288261	0.351944
qwen	0.066857	0.059048	0.222447	0.230952

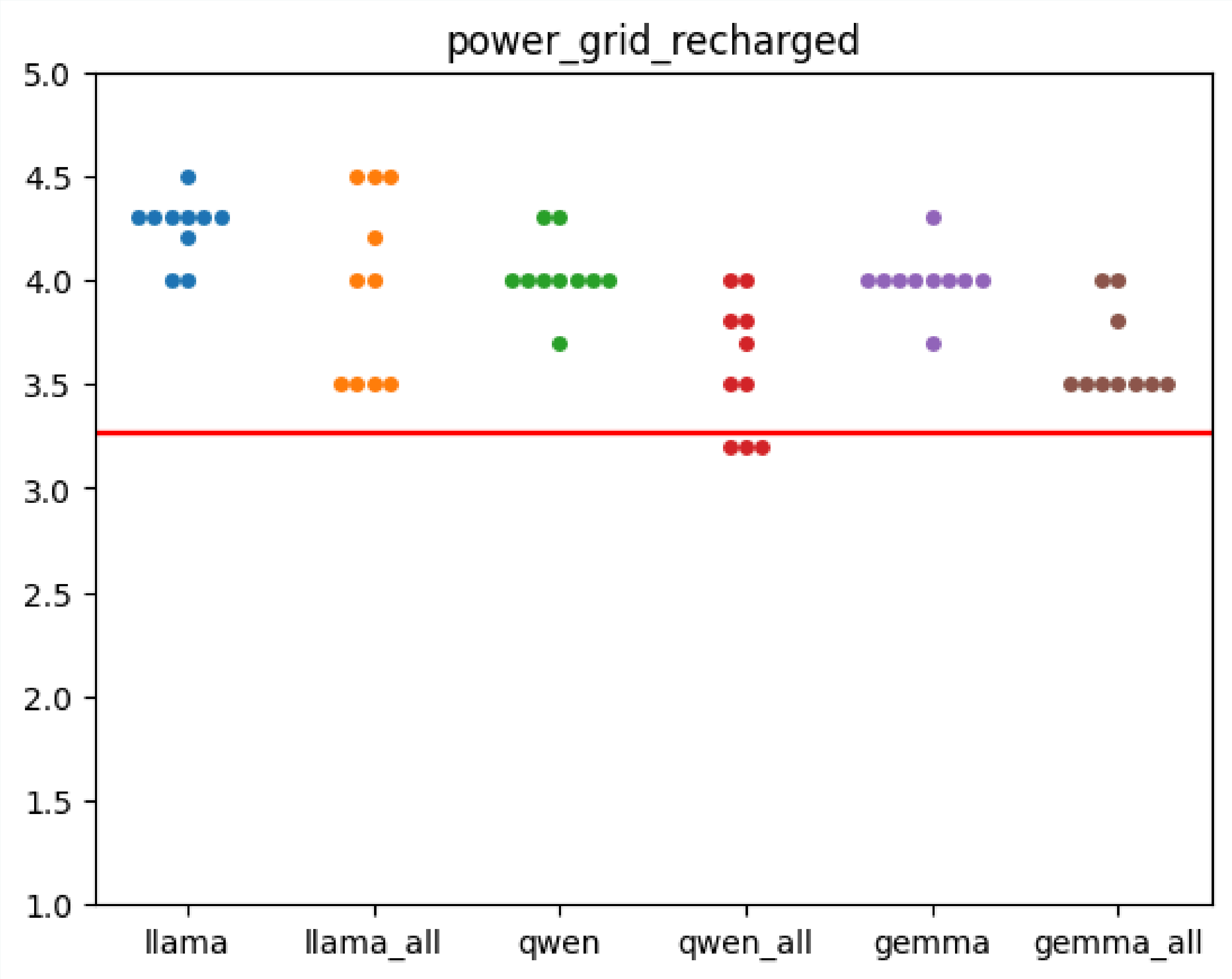
	Mean abs error		
Model	**Complexity**	**Optimal players**	**Duration**
llama	1.69	0.42	29.25
llama_all	1.23	1.05	33.00
qwen	1.22	0.30	32.40
qwen_all	1.06	0.58	24.38
gemma	1.09	0.33	20.38
gemma_all	1.05	0.48	35.88

Duration estimation is unreliable



Complexity overestimation





CONCLUSIONS

THANK YOU